

Data Dictionary — RPG Wheat Parcels

Master Thesis: Crop Harvesting Prediction · Winter wheat · Centre-Val de Loire, France

Source: Registre Parcellaire Graphique (RPG) — IGN / ASP · **Region:** Centre-Val de Loire (R24)
Editions: 2020–2024 (5 harvest seasons) · **Projection (raw):** Lambert-93 (EPSG:2154) · **License:** Open Licence v2.0 (Etalab)

1. Overview

The RPG is France's official Geographic Information System for agricultural parcels declared under the Common Agricultural Policy (CAP). Each parcel carries its boundary geometry and its declared main crop. This dictionary documents two layers: (1) the **raw RPG attributes** as delivered, and (2) the **processed parcel sample** produced for this project (1,500 wheat parcels with centroid coordinates).

Two tables, two purposes. The raw RPG (§2) is the source delivered by IGN — one row per agricultural parcel, all crops. The processed sample (§3) is the project deliverable — one row per sampled wheat parcel, enriched with a unique ID and latitude/longitude for downstream feature extraction (SoilGrids, Sentinel-2).

2. Raw RPG attributes (as delivered)

Granularity: one row = one agricultural parcel. Layer used: PARCELLES_GRAPHIQUES (shapefile, 2020–2023) / RPG_Parcelles (geopackage, 2024). Identical schema across all editions.

Field	Type	Unit	Description
ID_PARCEL KEY	int	—	Parcel identifier within an edition. Stable enough between consecutive years to track crop rotation, but treated as edition-specific in this project.
SURF_PARC RAW	float	hectares	Parcel area. Used to filter out small parcels (< 1 ha) for reliable satellite signal.
CODE_CULTU KEY	text	—	Main crop code. "BTH" = soft winter wheat (the project filter). ~350 crop codes exist in the full nomenclature.
CODE_GROUP RAW	text	—	Crop group code. "1" = soft wheat group. Coarser than CODE_CULTU.
CULTURE_D1 RAW	text	—	First catch/secondary crop, if any. Mostly empty for wheat parcels.
CULTURE_D2 RAW	text	—	Second catch/secondary crop, if any. Mostly empty.
geometry RAW	polygon	Lambert-93	Parcel boundary polygon. Used to compute the centroid.

2.1 Relevant crop codes (context)

Code	Crop (the filter target and common rotation crops)
BTH	Soft winter wheat — the project target crop
ORH	Winter barley (most common rotation after wheat)
CZH	Winter rapeseed
ORP	Spring barley
TRN	Sunflower
MIS	Maize

3. Processed parcel sample (project deliverable)

File: parcelles_ble_cv1_sample.csv / .geojson. Granularity: one row = one sampled wheat parcel for a given year. 1,500 rows (300 × 5 years). Coordinates in WGS84 (EPSG:4326).

KEY identifier **RAW** carried from RPG **DERIVED** computed for the project

Column	Type	Unit	Description
parcelle_uid KEY	text	—	Unique observation identifier , built as year_IDPARCEL (e.g. 2023_4253552). Primary key for joining soil and satellite features.
year KEY	int	year	Harvest year / RPG edition (2020–2024). Links the parcel to its campaign and target.
ID_PARCEL RAW	int	—	Original RPG parcel identifier (within its edition).
SURF_PARC RAW	float	hectares	Parcel area. All sampled parcels are > 1 ha (mean 7.9, max 55.9).
CODE_CULTU RAW	text	—	Crop code — always "BTH" in this sample (soft winter wheat).
lat DERIVED	float	degrees	Latitude of the parcel centroid (WGS84). Used to query SoilGrids and Sentinel-2.
lon DERIVED	float	degrees	Longitude of the parcel centroid (WGS84).

4. Sampling and processing summary

Step	Detail
Wheat filter	CODE_CULTU == "BTH"
Size filter	SURF_PARC > 1 ha
Sample size	300 parcels per year, 1,500 total
Sampling method	Random, fixed seed (42) for reproducibility
Per-year independence	Parcels sampled separately each year (crop rotation — see §5)
Centroid	Computed in Lambert-93 (metric), converted to WGS84 lat/lon

5. Data quality and structure notes

- **Crop rotation verified:** only 17.1% of 2020 wheat parcels were still wheat in 2021, confirming that per-year sampling is necessary (a fixed parcel set would mostly not be wheat in later years).
- **Repeated cross-section, not panel:** because parcels differ each year, the 1,500 rows are independent observations, not 300 parcels tracked over time.
- **Geographic control passed:** all centroids fall within Centre-Val de Loire (latitude 46.4–48.9, longitude 0.1–3.1).
- **Format consistency:** the 2024 GeoPackage exposes the same attributes as the 2020–2023 shapefiles, so all editions were processed identically.
- **Crop stability across years:** soft wheat is 12–16% of all parcels each year, consistent and realistic for this cereal region.

6. Project usage

- **Primary key:** `parcelle_uid` joins this sample to the soil features (SoilGrids) and satellite features (Sentinel-2), both keyed on the same parcels.
- **Coordinates:** `lat/lon` are the access keys to query external sources per parcel.
- **Target link:** `year` links each parcel to its regional harvest date (Céré’Obs) and its department’s weather (NASA POWER).
- **Reusability:** exported once and intended to be saved as a standalone dataset so all downstream notebooks share the identical parcel sample.